Aaron Leevord Fernandes

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TECHNICAL SKILLS

ML Modeling & Research: Reinforcement Learning (SAC, HER), Image Processing and Computer Vision (Segmentation, Registration), GenAI (Diffusion/Flow Matching/Rectified Flow, LLMs, VLMs)

Systems & Distributed Training: PyTorch (torch.profiler), DeepSpeed (ZeRO 1/2/3), DDP, Triton, Slurm (HPC), CUDA, MPI, Ray, JAX

Engineering & MLOps: AWS (SageMaker, EC2, S3), Docker, MLFlow, Langchain, FastAPI, vLLM, HuggingFace, Gymnasium

Languages & Data: Python (Expert), C++ (CUDA/Systems), Java (Spring Boot), SQL, PostgreSQL, Neo4j, NumPy, Pandas

EXPERIENCE

Machine Learning Research Engineer (Computer Vision) — Leung Research Group Apr 2024 – Present

- **Reduced FID by 50%** vs. Fourier-domain baselines by architecting an unpaired video-to-image (OCT-to-histology) translation pipeline using CNN-GRU and Transformer architectures in PyTorch.
- **Decreased simulation wall-clock time by 90%** by engineering scalable HPC data-generation pipelines (Slurm + MPI) for metasurface simulations, enabling 5K+ synthetic sample generation.
- **Accelerated metasurface design iteration** by training and evaluating neural surrogate models to approximate complex FDTD/FEM solvers.
- **Improved frame alignment accuracy** by using VoxelMorph-style diffeomorphic registration for temporal consistency in OCT video processing.

Machine Learning Engineer (RL) — Indiana University Jan 2025 – June 2025

- **Boosted task success rate from 15% to 55%** by implementing goal-conditioned SAC + HER and SVGG curriculum learning from scratch in PyTorch for sparse-reward MuJoCo environments.
- **Ensured 100% experiment reproducibility** by integrating MLflow for tracking 100+ training runs, including model versioning, checkpointing, and automated recovery.
- **Optimized training stability** through the implementation of modular replay buffers with scheduling, mini-batch updates, gradient clipping, and target network updates.

Software Engineer — Tata Consultancy Services (HDFC Bank) Jan 2022 – Jan 2023

- **Supported 100K+ daily transactions** by developing microservices-based transaction systems using Java, Spring Boot, and Hibernate in a production banking environment.
- **Collaborated in cross-functional teams** to maintain enterprise systems for a tier-1 bank, following strict CI/CD and production release protocols.

PROJECTS

Multimodal Image Generation (Rectified Flow) — *PyTorch, CLIP, DeepSpeed, ZeRO-2* *Nov 2025 – Jan 2026*

- **Developed a hybrid diffusion transformer based generative model** using Rectified Flow to improve sampling efficiency and prompt adherence over standard DDPM baselines.
- **Scaled training throughput** by implementing DeepSpeed (ZeRO-2) and DDP, optimizing GPU memory for massive scaling on HPC clusters.
- **Architected low-latency inference** via a FastAPI backend and Gradio UI, enabling real-time prompt-based image generation.

ComicPaliGemma (VLM Fine-tuning) — *SigLIP, Gemma, QLoRA, PyTorch* *Sept 2025 – Nov 2025*

- **Reduced peak VRAM requirements by 65%** via 4-bit QLoRA fine-tuning, enabling the training and optimization of a PaliGemma-based VLM on memory-constrained, single-GPU setups.
- **Engineered a multimodal synchronization pipeline** to align SigLIP ViT embeddings with Gemma autoregressive tokens, improving temporal and stylistic coherence in ”next-panel” comic sequence prediction.

LLM GraphRAG Analysis App — *Docling, Pinecone, Neo4j, LangChain, Electron* *Oct 2024 – Dec 2024*

- **Architected a GraphRAG pipeline** utilizing **Docling** for document layout parsing and **Sentence Transformers** for Pinecone indexing; reduced research timelines from months to days with 90% human-code alignment.
- **Engineered a hybrid retrieval system** by mapping semantic vector embeddings (Pinecone) to structured Knowledge Graphs (**Neo4j**), enabling LLMs to navigate complex thematic hierarchies and multi-hop relationships.
- **Deployed a cross-platform desktop suite** via Electron.js and D3.js, providing researchers with interactive visualizations of LLM-generated codes and real-time relationship mapping within the Grounded Theory framework.

Unsupervised Industrial Defect Detection — *ViT, Mixture Density Networks, PyTorch* *Jan 2025 – May 2025*

- **Achieved 0.94 AUROC** by designing a ViT-B/16 autoencoder with a 10-component Mixture Density Network (MDN) to model latent distribution density, outperforming ResNet-50 reconstruction baselines.

EDUCATION

Indiana University Bloomington — M.S. in Intelligent Systems Engineering May 2025

- **GPA: 3.9/4.0** — Coursework: Image Processing, Deep Learning Systems, Reinforcement Learning, NLP, Cloud Computing, Multivariate Statistics.

Goa University — B.E. in Electronics and Telecommunication July 2021